CMPE 256 Recommender Systems – Term Project Guidelines

1. Project Overview

The purpose of this capstone project is to bridge the gap between theory and practice. Working in **teams of 3**, you will design, implement, and evaluate a fully functional Recommender System. You are expected to identify a specific domain (e.g., e-commerce, finance, education, real estate, social media, jobs, food), select a suitable public dataset (e.g., Kaggle, UCI, MovieLens), and build a solution that addresses a real-world problem.

Key Expectation: While using standard libraries is allowed, **novelty and originality are highly encouraged**. Projects that go beyond simple "out-of-the-box" implementations (e.g., by creating hybrid models, incorporating novel methodologies/techniques, using Large Language Models and Agentic AI) will receive higher grades.

---

2. Project Phases & Deliverables

Phase 1: Project Proposal (Mandatory)

- **Action:** Submit a 1-page PDF proposal for instructor approval before writing any code.
- **Content:**
    - **Team:** Names of all 3 members.
    - **Objective:** What problem are you solving?
    - **Dataset:** URL to the dataset (Must be sizable, e.g., >50k interactions).
    - **Methodology:** Which baseline and advanced models do you plan to use?
    - **Novelty Factor:** What is unique about your approach?

Phase 2: Implementation & Analysis

- **Development:** Data cleaning, EDA, Model training, Hyperparameter tuning.
- **Interface:** Create a simple frontend (Streamlit, Flask, or CLI) to demonstrate the recommendations.

Phase 3: Final Presentation & Demo (Final Week)

- **Format:** 15-minute Oral Presentation + 5-minute Q&A.
- **Requirement:** All team members must present.
- **Demo:** A live demonstration of the system working (e.g., input a user ID, receive recommendations).

Phase 4: Final Submission

- **Submission Platform:** Canvas.
- **Deliverables:**
    1. Presentation Slides (PPT/PDF).
    2. Final Project Report (PDF).
    3. GitHub Repository Link (included in the report).

---

3. Final Report Structure

The report should be a comprehensive technical document (approx. 10–15 pages).

1. **Project Goals:** Define the business value. Why does this recommender system matter?
2. **Data Description:** Source, sparsity analysis, and Exploratory Data Analysis (EDA) visualizations.
3. **Methodology (Novelty):**
    - Detail your algorithms (Collaborative Filtering, Content-Based, Hybrid, Deep Learning, etc.).
    - **Highlight your original contribution.** Did you engineer new features? Did you use a specific loss function? Did you integrate an LLM?
4. **Benchmark & Evaluation:**
    - Compare your model against a Baseline (e.g., Most Popular) and a standard model (e.g., KNN).
    - Use appropriate metrics (RMSE for ratings; Precision@k, Recall@k, NDCG for ranking).
5. **Lessons Learned & Takeaways:** Challenges faced (Cold Start, Scalability) and how you solved them.
6. **Source Code:** Link to GitHub.
7. **Contribution Table:** A detailed breakdown of who did what (e.g., *Member A: Backend & API*, *Member B: Data Cleaning & EDA*, *Member C: Model Training & Slides*).

---

4. GitHub Guidelines

- **Read Me (README.md):** Must be detailed. It should allow anyone to clone the repo and run the recommender.
    - Includes: Project Title, Setup Instructions (pip install -r requirements.txt), and Usage Guide.
- **Code Quality:** Modular code, comments, and organized folder structure.

---

5. Evaluation Rubric (Total: 100 Points)

The grading will be strict on functionality and generous on creativity.

Category	Weight	Criteria for High Marks
Novelty & Originality	**20%**	The project goes beyond basic library implementation. Evidence of creative feature engineering, hybrid architectures, unique domain selection, or integration of advanced tech (LLMs/Graph Neural Networks). The team attempted something difficult rather than the easiest path.
Oral Presentation & Demo	**20%**	**Live Demo works flawlessly.** Presentation is engaging, visual, and strictly adheres to time limits. All team members speak. The team answers Q&A questions confidently and correctly.
Technical Implementation	**25%**	Code works and is reproducible. Advanced algorithms are implemented correctly. The system handles edge cases (e.g., what happens if a new user enters?). Code quality is high (clean, commented).
Evaluation & Benchmarking	**15%**	Rigorous evaluation. The team understands *why* they chose specific metrics (e.g., using NDCG for ranking). A clear comparison between the proposed model and a baseline is shown in tables/graphs.
Report Quality & EDA	**10%**	Report is professional, well-structured, and free of typos. EDA provides deep insights into the data (not just basic histograms). The logic flows clearly from problem to solution.
GitHub & Documentation	**10%**	Repository is public and organized. README is comprehensive (includes setup/run instructions). requirements.txt is present. Contribution table confirms equal work distribution.

Important Note: If the live demo fails completely during the presentation, the maximum score for the "Presentation & Demo" section will be capped at 50%. Always have a backup video!

Here is the specific **Deliverables** section to be added at the end of your project guidelines document.

---

6. Project Deliverables Checklist

To ensure a smooth grading process, every team is required to submit the following items by the deadline. Failure to submit any of these components will result in a deduction of points.

A. Canvas Submission (One submission per team)

1. **Final Project Report (PDF)**
    - **Format:** Single PDF document.
    - **Naming Convention:** TeamName_FinalReport.pdf
    - **Content:** Must include all sections listed in Section 3 (Goals, Data, Methodology, Evaluation, Lessons Learned, Contribution Table).
2. **Presentation Slides (PPTX or PDF)**
    - **Content:** The exact slides used during your oral presentation.
    - **Naming Convention:** TeamName_Slides.pdf
3. **GitHub Repository Link**
    - Submit the URL to your public repository in the Canvas comment section **and** include it as a hyperlink in your Final Report.

B. GitHub Repository (Source Code)

Your repository must be public and include the following:

1. **Source Code:** Jupyter Notebooks (.ipynb) or Python scripts (.py).
2. **README.md:** A professional landing page for your project. It must include:
    - Project Title and Team Members.
    - **"How to Run":** Step-by-step instructions to install dependencies and run the recommender.
    - **Dataset Link:** If the dataset is too large for GitHub (>100MB), provide a link to the source (e.g., Kaggle/Google Drive) in the README. Do not upload massive CSV files directly.
3. **requirements.txt:** A list of all necessary libraries (e.g., pandas, scikit-surprise, pytorch) to ensure your code is reproducible.

C. In-Class Deliverable (Final Week)

1. **Oral Presentation:**
    - **Time Limit:** 15 Minutes Presentation + 5 Minutes Q&A.
    - **Requirement:** All 3 team members must speak.
2. **Live System Demo:**
    - A real-time demonstration of the recommendation engine (e.g., showing a web interface, a CLI interaction, or a notebook execution).
    - **Backup Plan:** It is **highly recommended** to record a 2-minute video of your demo working and have it ready on a USB drive or cloud link, just in case the live demo fails due to technical issues.